

COMSATS University Islamabad, Abbottabad Campus



PROJECT PROPOSAL

for

FixKar — (Local Handyman Booking Application)

Version 1.0

Submitted by

Team Member	Registration No.
Umer Saleem	SP24-BSE-171
Muzamil Naveed	SP24-BSE-150
Saad Bin Zahid	SP24-BSE-153
Sami Ullah	SP24-BSE-156

Supervisor: Sir Ahmed Ali

Bachelor of Science in Computer Science (2022–2026)

1. Introduction

Finding a reliable, trustworthy handyman — an electrician, plumber, painter, AC technician, carpenter, mechanic, or cleaner — is still largely an informal, word-of-mouth process in most local markets. Customers rely on unverified classifieds or personal referrals, with no guarantee of the provider's identity, skill, or pricing. FixKar is proposed as a two-sided mobile marketplace that connects Customers directly with verified local Service Providers, replacing this informal process with a structured, transparent, and accountable digital platform.

2. Problem Statement

Customers currently have no reliable way to verify a handyman's identity, skill level, or pricing before hiring them, which leads to fraud risk, no-shows, inconsistent service quality, and disputes over cost. At the same time, skilled Service Providers lack a digital channel to reach a steady stream of customers and manage bookings, payments, and their reputation in an organized way. FixKar addresses both sides of this gap through a single connected platform.

3. Project Objectives

- **Verified Marketplace:** Provide a platform where every Service Provider is identity-verified by an Admin before they can accept bookings.
- **Simple Booking Experience:** Let Customers discover, book, and track a service provider in a few taps, with real-time status updates.
- **Secure Payments:** Support tokenized online payment and cash-on-service, with automatic commission handling for the platform.
- **Trust and Accountability:** Build a rating and review system, plus a complaint-resolution workflow, to keep quality high on both sides.
- **Platform Governance:** Give FixKar administrators the tools to manage categories, verify providers, and monitor the platform through a web-based Admin Panel.

4. Proposed System Overview

FixKar consists of three cooperating parts sharing a common backend: a Customer-facing mobile application, a Service-Provider-facing mobile application, and a web-based Admin Panel. Customers browse services by category, view a provider's verified profile, and request a booking for a preferred date, time, and address. The request is matched to nearby, verified providers, who can accept or reject it within a configurable response window. Once accepted, the Customer can track the booking's status in real time — including the provider's live location — through to completion, followed by payment and an optional rating and review. The table below summarizes the system's core modules.

Module	What It Does
Authentication Module	OTP-based registration, secure login, role-based access for Customer, Provider, and Admin.
Customer Module	Service browsing, provider discovery, booking requests, live tracking, ratings and reviews.
Service Provider Module	Profile creation, document verification, booking accept/reject, job-status updates, earnings tracking.
Admin Module	Provider verification, category management, booking/payment oversight, complaint resolution, reporting.
Booking Module	End-to-end booking lifecycle: creation, provider matching, status tracking, and cancellation handling.
Payment Module	Secure online payments and cash-on-service handling with automated commission calculation.
Location Module	Google Maps-based address capture, proximity matching, and live technician tracking.
Notification Module	Real-time push and in-app alerts via Firebase Cloud Messaging.

5. Scope

5.1 Within Scope (Version 1.0)

- Customer and Service Provider registration with OTP-based mobile/email verification.
- Service discovery, category browsing, and keyword/filter search.
- Booking creation, provider matching, and real-time status tracking with Google Maps location support.
- Secure online and cash payments with automatic commission calculation.
- Push notifications, ratings and reviews, and complaint management.
- Web-based Admin Panel for provider verification, category management, and reporting.

5.2 Out of Scope (Version 1.0)

- Multi-country or multi-currency operation.
- Provider payroll or tax computation beyond commission deduction.
- A fully automated dispute-arbitration workflow beyond Admin-mediated resolution.
- Integration with third-party home-insurance or warranty providers.

6. Technology Stack

- **Mobile Applications:** Built with Flutter (Dart) for a single codebase that runs on Android now and can extend to iOS later.
- **Backend:** Firebase — Authentication, Cloud Firestore, Storage, and Cloud Messaging — removing the need to provision or maintain physical servers.
- **Location Services:** Google Maps Platform API for geocoding, proximity ranking, and live tracking.
- **Payments:** A licensed third-party Payment Gateway using tokenized transactions; FixKar never stores raw card data.
- **Admin Panel:** A responsive web application for desktop browsers (Chrome, Edge, Firefox).

7. Development Methodology and Timeline

The project will be developed using an Agile, sprint-based approach across an indicative 18-week schedule, allowing the four-member team to deliver working increments of the system and incorporate supervisor feedback throughout the semester.

Phase	Planned Work
Sprint 1 (Weeks 1-3)	Authentication Module, Flutter project setup, Firebase configuration.
Sprint 2 (Weeks 4-6)	Customer browsing/search, Service Provider profile & verification upload.
Sprint 3 (Weeks 7-9)	Booking Module and Location Module integration.
Sprint 4 (Weeks 10-12)	Payment Module and Notification Module.
Sprint 5 (Weeks 13-15)	Review & Rating, Complaint Module, Admin Module core screens.
Sprint 6 (Weeks 16-18)	Admin reports/dashboard, end-to-end testing, and documentation.

8. Expected Outcome

By the end of the project, FixKar will deliver a functioning two-sided mobile marketplace — Customer app, Service Provider app, and Admin Panel — capable of supporting verified provider onboarding, end-to-end booking and payment flows, real-time tracking, and platform governance. The system is designed to scale to at least 10,000 registered Customers and 1,000 verified Service Providers without architectural change, and lays the groundwork for future enhancements such as an iOS release and Urdu localization.

9. Conclusion

FixKar offers a practical, technically sound solution to a real, everyday problem: finding a handyman you can trust. By combining identity verification, transparent pricing, real-time tracking, and accountable ratings within a single Flutter and Firebase-based platform, the project aims to reduce fraud and no-show risk for Customers while giving Service Providers a steady, digitally managed source of work. Full functional and non-functional requirements, system models, and database design are documented in the accompanying Software Requirements Specification (SRS).